Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

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Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

_show ip route ospf____

Part 4: Configure OSPF Passive Interfaces

g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? ___Serial0/0/0__________

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? __129_______

Does R2 show up as an OSPF neighbor on R1? __Yes______

Does R2 show up as an OSPF neighbor on R3? __No______

What does this information tell you?

____R2 only has OSPF advertising enabled on its S0/0/0 interface and has only established an OSPF adjacency with R1. R2's S0/0/1 interface remains a passive interface, therefore, R3 can only forward traffic to 192.168.2.0/24 through R1, and it is not adjacent to R2.________________________________

h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

___router ospf 1____

___no passive-interface s0/0/1____

i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? __Serial0/0/1__

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

The metric is 65; OSPF cost of R3's S0/0/1 interface (64) + OSPF cost of R2's G0/0 interface (1) = 65

Is R2 listed as an OSPF neighbor to R3? ___Yes_____

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
R1(config-router)# auto-cost reference-bandwidth 100
% OSPF: Reference bandwidth is changed.
        Please ensure reference bandwidth is consistent across all routers.
```

Why would you want to change the OSPF default reference-bandwidth?

OSPF's default reference bandwidth is 100Mbps. All links with a bandwidth of $\geqslant$100Mbps will have a calculated cost of 1, making it impossible to distinguish the bandwidth differences between different high-speed links. Modifying the reference bandwidth allows OSPF to calculate a more accurate cost value based on the actual link bandwidth, making the route selection more in line with the actual network bandwidth performance.

Step 2: Change the bandwidth for an interface.

g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

192.168.3.0/24: After modifying the bandwidth of R1's S0/0/1 interface, the cost is 781 + the default cost of R3's G0/0 interface is 1 = 782;

192.168.23.0/30: The cost of R1's S0/0/1 interface is 781 + the default cost of R3's S0/0/1 interface is 64 = 845 (both the two equivalent paths S0/0/0 and S0/0/1 have a cost of 845).

i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

The new cumulative cost is 1562; All serial interfaces have been modified to 128Kbps, with each serial link costing 781; the path to 192.168.23.0/30 is either R1-S0/0/0-R2-S0/0/1 or R1-S0/0/1-R3-S0/0/1, both being two serial links, totaling 781+781=1562, thus forming two equivalent paths.

Step 3: Change the route cost.

c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

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_____After manually configuring IP OSPF cost 1565 for the S0/0/1 interface of R1, the cost of a single segment of the S0/0/1 interface directly connected to R3 is already higher than the cumulative cost from R2 to R3; OSPF uses the Shortest Path First algorithm, which will select the path via R2 with the lower cost as the optimal route.______________________________________

Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

   _The Router ID is a unique identifier for routers within an OSPF domain. Adjacency establishment, LSA switching, and SPF algorithm calculation all rely on the Router ID. Without manual control, the Router ID is taken from the highest IP address of the router's active physical interface. Interface failures or IP changes will cause the Router ID to change, leading to OSPF adjacency interruptions, routing information re-flooding, and network convergence fluctuations. Manually specifying the Router ID ensures identifier stability and facilitates network management and troubleshooting.____

2. Why is the DR/BDR election process not a concern in this lab?

   _All serial links in this experiment are point-to-point networks. OSPF does not perform DR/BDR election on point-to-point links; it directly establishes adjacency relationships. Only broadcast multiple access (such as Ethernet) networks require DR/BDR to reduce LSA flooding. Therefore, DR/BDR election is not a concern in this experiment.____

3. Why would you want to set an OSPF interface to passive?

   _Setting an interface to passive prevents OSPF from sending Hello messages on that interface, avoiding the establishment of unnecessary neighbor relationships, while still advertising the network where that interface resides to other OSPF routers. This is typically used in end-user networks to save bandwidth, improve security, and reduce unnecessary protocol overhead.____

4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

   _OSPF supports a multi-area design. By dividing the network into a backbone area (Area 0) and multiple regular areas, each area maintains its own link-state database, while areas exchange information through summarized routes. This hierarchical structure reduces the size of routing tables and limits the flooding range of LSAs, thereby improving network scalability.____

5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

   ____network 10.0.0.0 0.255.255.255 area 0____

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6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

   _EIGRP routes will be selected. Administrative distance takes precedence over metric values. The default AD values for different routing protocols are: EIGRP (internal) = 90, OSPF = 110, and RIP = 120. The smaller the AD value, the higher the protocol priority. Therefore, even if the EIGRP metric value is larger, it will still be selected as the optimal route.

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